

REPUBLIC OF THE PHILIPPINES
Laguna State Polytechnic University
ITEP 414 – System Administration and Maintenance

WEEK 2 PORTFOLIO PROJECT

Enterprise Infrastructure Planning for a Startup Company

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Program: Bachelor of Science in Information Technology
Project Type: Individual Portfolio Project
Week: 2

ABC Startup Solutions

PART 1 – COMPANY PROFILE

Company Name

ABC Startup Solutions

Nature of Business

ABC Startup Solutions is a newly established software development company that provides customized software applications, web-based systems, and digital technology solutions for small and medium-sized businesses. The company focuses on developing reliable, user-friendly, and cost-effective software solutions that help organizations improve daily operations, productivity, and business processes.

The company operates with four main departments: Information Technology, Human Resources, Finance, and Sales. These departments work together to support software development, employee management, financial operations, and business development.

Company Vision

To become a trusted technology solutions provider that helps businesses grow through innovative, reliable, and accessible digital solutions.

Company Mission

ABC Startup Solutions is committed to developing quality software and digital solutions that address the needs of businesses. The company aims to provide reliable technology services, maintain secure and efficient IT operations, and continuously improve its products and services through innovation and professional teamwork.

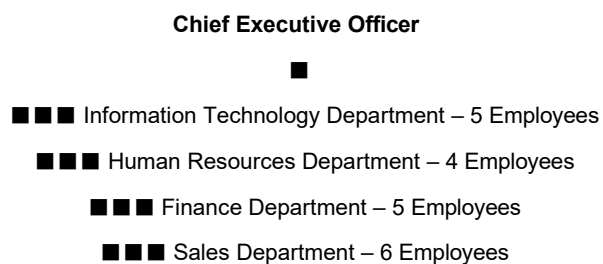
Office Location

Fictional Office Location: 2nd Floor, ABC Business Center, National Highway, Barangay San Isidro, Santa Rosa, Laguna, Philippines.

The company occupies a single office floor where all four departments operate. The centralized office setup allows employees and departments to communicate efficiently while sharing common IT infrastructure and network resources.

Organizational Structure

ABC Startup Solutions follows a functional organizational structure in which employees are grouped according to their responsibilities and areas of expertise.



Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

PART 2 – ENTERPRISE HARDWARE INVENTORY

The hardware inventory is designed for the company's 20 employees and single-floor office. The quantities prioritize one primary workstation per employee while providing shared infrastructure, backup, and network equipment.

Asset ID	Hardware	Qty.	Department	Purpose
HW-001	Desktop Computers	16	IT, HR, Finance, Sales	Primary workstation for office-based employees.
HW-002	Laptops	4	IT, Sales	Portable workstations for IT staff, meetings, and presentations.
HW-003	Server	1	IT	Hosts internal services, shared files, and selected applications.
HW-004	Router	1	IT	Routes traffic between the LAN and the ISP connection.
HW-005	Managed Switch	1	IT	Provides centralized wired LAN connectivity.
HW-006	Network Printer	2	Shared	Provides shared printing for departments.
HW-007	UPS	2	IT / Server Room	Protects server and core network equipment from power interruption.
HW-008	Wireless Access Point	2	Shared	Provides secure wireless coverage across the office.
HW-009	NAS Storage	1	IT	Centralized file storage and secondary backup destination.
HW-010	External Backup Drive	2	IT	Supports offline backup rotation and recovery.
HW-011	Monitors	20	All Departments	One primary display for each employee workstation.

PART 3 – ENTERPRISE SOFTWARE INVENTORY

The software inventory covers the required operating systems, productivity, development, version-control, security, remote-support, browser, virtualization, and file-compression tools.

Software	Version	License	Purpose
Windows 11 Pro	25H2	Commercial	Primary OS for employee desktops and laptops.
Ubuntu Server	24.04 LTS	Open Source	Server OS for internal services and server workloads.
Microsoft 365 Apps	Current Channel	Subscription	Documents, spreadsheets, presentations, and productivity.
Visual Studio Code	Latest Stable	Free / Open Source	Code editor for software development and administration tasks.
Git	Latest Stable	Open Source	Version control for source code and configuration files.
GitHub Desktop	Latest Stable	Free	Graphical Git workflow for repository management.
VirtualBox	7.x	Open Source / Free	Virtualization for testing operating systems and lab environments.
Google Chrome	Latest Stable	Free	Web browser for business applications and web access.
Microsoft Defender	Built-in	Included with Windows	Endpoint malware and threat protection.
AnyDesk	Current Version	Commercial / Free options	Remote support and troubleshooting when authorized.
7-Zip	24.x	Open Source	File compression, extraction, and archive management.

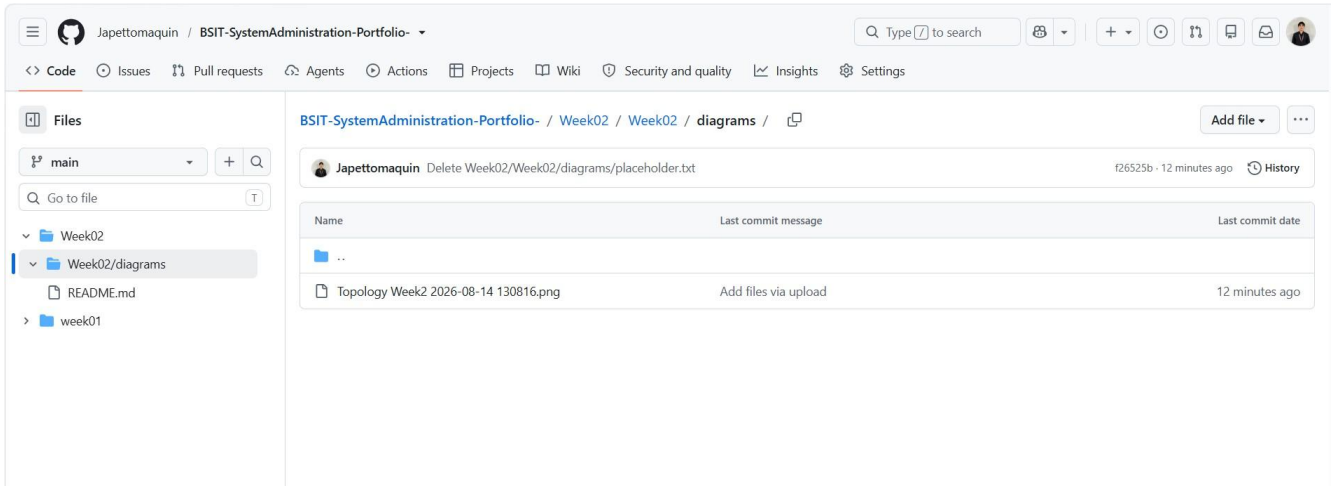
PART 4 – ENTERPRISE NETWORK INVENTORY

The network inventory supports the required enterprise topology and structured cabling for a 20-person office.

Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modem / ONT	1	Connects the office to the internet service provider.
NET-002	Router	1	Performs routing, NAT, and basic network services.
NET-003	Firewall	1	Filters traffic and protects the internal network.
NET-004	Managed Switch	1	Connects wired devices and supports network segmentation.
NET-005	Wireless Access Point	2	Provides secure Wi-Fi connectivity.
NET-006	Patch Panel	1	Organizes and terminates structured network cabling.
NET-007	CAT6 Cable	300 m	Structured cabling for workstations and network equipment.
NET-008	RJ45 Connectors	50	Terminates Ethernet cables and supports installation/maintenance.

PART 5 – ENTERPRISE NETWORK DIAGRAM

The following is the network topology image to be used for Part 5.



PART 6 – SYSTEM ADMINISTRATION ROLES

Helpdesk Technician

Responsibilities: Provides first-line technical support; receives and documents user incidents; installs and configures approved software; assists with account and workstation issues; escalates complex problems to the appropriate administrator.

Skills: Hardware and software troubleshooting, Windows administration, basic networking, ticket documentation, customer service, communication, and problem-solving.

Common Tools: Ticketing/helpdesk system, remote-support software, Windows administrative tools, device diagnostics, and basic network utilities.

Certifications: CompTIA A+ is a suitable entry-level certification for demonstrating foundational IT support knowledge.

Network Administrator

Responsibilities: Designs, configures, monitors, maintains, and troubleshoots the company's LAN, router, firewall, switches, and wireless infrastructure; manages IP addressing and network access.

Skills: TCP/IP, subnetting, VLANs, routing, switching, wireless networking, network security, monitoring, and troubleshooting.

Common Tools: Cisco IOS or equivalent network CLI, network monitoring tools, packet capture tools, ping/traceroute, cable testers, and configuration backups.

Certifications: Cisco CCNA is a relevant certification covering networking fundamentals, IP connectivity, services, security fundamentals, and automation.

Linux System Administrator

Responsibilities: Installs, configures, secures, updates, monitors, and troubleshoots Linux servers; manages users, permissions, storage, services, logs, and backups.

Skills: Linux command line, filesystems, users and groups, permissions, networking, service management, shell scripting, logging, and troubleshooting.

Common Tools: Ubuntu Server, SSH, systemd, journalctl, package managers, shell utilities, monitoring tools, and backup utilities.

Certifications: Linux Foundation Certified System Administrator (LFCS) is a relevant vendor-neutral certification focused on hands-on Linux administration.

Cloud Administrator

Responsibilities: Manages cloud resources, identities, storage, networking, monitoring, security controls, backups, and cloud costs; supports migration and integration between on-premises and cloud services.

Skills: Cloud fundamentals, IAM, virtual networks, compute, storage, monitoring, security, automation, backup, and cost management.

Common Tools: AWS/Azure/GCP management consoles and CLIs, identity and access tools, monitoring services, infrastructure-as-code tools, and backup services.

Certifications: AWS Certified Cloud Practitioner is an appropriate foundational certification for demonstrating knowledge of AWS Cloud concepts, services, security, and billing.

Collaboration

The four professionals work together as a coordinated IT support and infrastructure team. The Helpdesk Technician serves as the first point of contact for employee issues, documents incidents, resolves common workstation problems, and escalates complex cases. The Network Administrator maintains the connectivity, routing, switching, wireless, and network-security components that allow users and systems to communicate. The Linux System Administrator manages

PART 7 – INFRASTRUCTURE RECOMMENDATIONS

Internet Provider

For the initial design, a business-grade fiber connection is recommended. A practical example is Globe Business GFiber Biz, with a plan capable of supporting a 20-person office. The exact subscription should be selected after checking service availability, business support, SLA terms, and expected bandwidth.

Server Specifications

Recommended baseline: a business-class tower or rack server with a modern 8-core processor, 32 GB ECC RAM, 2 × 1 TB SSDs configured as RAID 1 for the operating system and applications, additional storage for shared data or a separate NAS, dual network interfaces where supported, redundant power supply if budget permits, and Ubuntu Server 24.04 LTS.

Backup Strategy

Use a 3-2-1 backup approach: three copies of important data, on at least two different storage media, with at least one copy offline or off-site. Use daily incremental backups, scheduled full backups, NAS storage, rotating external drives, encryption, monitoring, and periodic restore tests.

Security Recommendations

Use a perimeter firewall, strong administrative credentials, multi-factor authentication where supported, network segmentation, regular patching, least-privilege access, secure Wi-Fi, centralized logs where practical, and documented incident-response and acceptable-use procedures.

Antivirus

Use Microsoft Defender on Windows 11 Pro endpoints and keep protection features, security intelligence updates, and operating-system updates enabled. Combine endpoint protection with patching, firewall controls, least privilege, user awareness, and backups.

Password Policy

Require unique passwords, prohibit sharing, use a password manager for authorized administrative credentials, enable multi-factor authentication for important services, separate administrative accounts from ordinary accounts, and disable accounts immediately when employees leave.

Expansion Plan

Keep spare switch ports and rack capacity, document and label CAT6 cabling, reserve IP address space, and plan for additional access points, switches, storage, server resources, cloud services, and bandwidth as the company grows. Consider a secondary Internet connection when downtime becomes a significant business risk.

PART 8 – PERSONAL REFLECTION

This project taught me that system administration does not begin with computer installation and network equipment deployment. It helped me learn how to evaluate the needs of a small business and identify requirements in terms of hardware, software, network infrastructure, and equipment. In this project, I also realized that compiling inventories assists in understanding what resources should be allocated to particular assets, their quantities, and responsible departments. The network diagram helped me see how Internet access, routing and firewall security, switches, wireless access points, servers, printers, and employees' departmental affiliation fit together.

The most challenging aspect of this project was determining what equipment and specifications to recommend based on the given limitations of a 20-employee start-up. It was difficult to prioritize options and select the most optimal ones without making the recommendations sound too expensive. Another valuable aspect of this project was researching what system administration positions were available and understanding that no single administrator manages all IT infrastructure aspects. This project helped me realize the importance of collaboration between helpdesk, network, Linux, and cloud administrators.

I think planning is essential before doing anything because mistakes at the beginning of the process can be costly and time-consuming to fix. A thorough project plan helps with budgeting, purchasing, configuring, securing, documenting, troubleshooting, and even expanding the infrastructure. In other words, it guides the team through the implementation and ensures that the company knows why it spends the money on specific technologies.

I believe this project will benefit me later because it gives me the opportunity to gain experience working on various aspects of system administration. I can develop my planning, inventory compiling, network design, technical documentation, security design, and technical decision-making skills. Moreover, I will always remember that a system administrator's role is not to focus on individual components but rather on the entire system. In other words, my role is to ensure that all elements support business continuity, be reliable, secure, scalable, and not waste the company's resources.

REFERENCES

- Laguna State Polytechnic University. ITEP 414 – System Administration and Maintenance, Week 2 Portfolio Project: Enterprise Infrastructure Planning for a Startup Company.
- Cisco. Network Administrator and CCNA certification information.
- Linux Foundation. Linux Foundation Certified System Administrator (LFCS).
- AWS. AWS Certified Cloud Practitioner (CLF-C02).
- Globe Business. GFiber Biz business broadband information.